engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation *(G1)*

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in-between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date: 3/10/24
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:
Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

Prototype 1: VEX Door

	Details	Tools and Manufacturing Methods used
Step 1	Frame	VEX Materials (Metal frame, nuts and bolts, chain, standoffs)

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Step 2	Attach Gate into the frame	Blast Gate
Step 3	Secure Gate	VEX Screws and nuts
Step 4	Attach Chain onto gears	VEX sprockets, axles, chain
Step 5	Attach gate's openable door to chain	VEX screws, standoffs, electrical tape
Step 6	Signal for status display, rotatable signal that displays either OPEN in green, or CLOSED in red	VEX gear, small metal plates, tape and markets

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Prototype 2: Cable Action Gate

Date: 5/29/24
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:
Details about all steps of construction (add step numbers as needed), including the tools

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and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Use CNC to cut lever parts out of wood	CNC Machine, board of wood
Step 2	Cut a groove into the circular center piece of wood	Wood mill
Step 3	Attach bolts to the circular centerpiece, cutting holes and gluing VEX pieces to make clamps for bike wire	Drill press, VEX standoffs, screws, superglue
Step 4	Create and assemble custom blast gate	3D printer, CNC machine, wood
Step 5	Run bike wire through exterior white cable, attach through blast gate	Bike Wire
Step 6	Seal a cap onto the other side of the blast gate	Soldering Iron
Step 7	Assemble wood lever	Wood glue, nailgun
Step 8	Attach bike wire to clamps, run through wood lever holes	Screwdriver

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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	**Description**	**Where material was purchased**	**Cost per unit**	**# units**	**Total cost per item**
VEX Standoff	A small piece of VEX material that adds space between 2 screws	Gregory owned	$0.8	2	$1.6
VEX Screw	A screw for VEX materials	Gregory Owned	$0.15	20	$3
VEX Nut	A nut	Gregory Owned	$0.32	20	$6.4
VEX Sprockets	A gear that has teeth for holding chains	Gregory Owned	$1.7	1	$1.7
VEX Chain	Chain, compatible with sprockets and screws	Gregory Owned	$0.65 / ft	1	$0.65

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VEX Metal pieces	Screwable metal plates and building blocks for VEX, comes in different shapes and sizes	Gregory Owned	Roughly $5.5 each	8	$44
Sample Blast Gate	A blast gate, attachable to a hose to open and close	[Home Depot]()	$8.05	1	$8.05
Plywood	A large 4ft x 8ft, cut down to cutting size	Gregory Owned	~$20	0.5	~$10
3D Printing PLA	Roll of PLA for 3D printing	Gregory Owned	~$8	0.1 (For the print)	$0.8
Bike Wire / Cable	An exterior white cable with an interior metal one, used for the blast gate	Amazon	$13	1	$13

Detailed sketches, which might include CAD if you have learned it in your class:

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Construction of the Final Prototype *(G2)*

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Able to fully open and close in 5 seconds	We will be able to see if the gate will be efficient and functional by testing how long it takes to open the gate, which hopefully will be swift and quick
Can the gate open fully with one turn of the lever	We will be able to see if there are any problems or disconnect between the intended function of the lever, and the opening or closing of the blast gate
Intuitively understood by 3 students	By asking 3 students to use the apparatus, we will be able to see if students (who also use the workshop) will intuitively understand the design

Testing or Modeling of Attributes *(G3, G4)* and Addressing Non-Testable Attributes *(G4)*

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

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For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	**Description**	**Describe your prototype's functionality to test this attribute**
Visual recognizability	Will students be able to see what the design is and does?	Students should be able to see that it's a lever and that it would connect to the gate
Visual strikingness	Will students be able to recognize the design among the workshop equipment?	Students should know to use it and find it in the shop
Ease of use	Will students be able to easily use the design repeatedly?	In theory it should be able to work

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Addressing Non-Testable Attributes *(G4)*

For all attributes that you listed “expert review required” in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	**Justification for why this attribute cannot be modeled or tested with your prototype**
Recognizability	Subjective, based on visuals and how students view the world
Strikingness	Based on the workshop and again, subjective visuals
Ease of use	Dependent on how well individual students will understand the mechanism and how to use it